# CALCULATING INPUT VALUES

Knowns:

- Gear Ratio (GR) = 24:9 = 24/9
- $N_{out}$ = 1100 RPM (Speed of output)
- $T_{out}$ = 1.25 in-lbf (Torque output)

Unknowns:

- $N_{in}$ (Speed of input)
- $T_{in}$ (Torque input)
- Power (P)

## Calculating Input Speed ($N_{in}$):

From Machinery's Handbook: (M.H)

$$N_{in} = N_{out}(GR) \quad (eq\ 1)$$

$$N_{in} = (1100\ RPM)\left(\frac{24}{9}\right)$$

$$\boxed{N_{in} = 2933.33\ RPM}$$

## Calculating Input Torque ($T_{in}$):

From M.H: $T_{out} = T_{in}(GR)$ (Eq 2)

$$T_{in} = \frac{T_{out}}{GR} = \frac{(1.25\ in\cdot lbf)}{(24/9)}$$

$$\boxed{T_{in} = 0.46875\ in\cdot lbf}$$

## Calculating Power (P):

$$P_{out} = P_{in} = P = \frac{T_{out}\cdot N_{out}}{63000}$$

$$P = \frac{(1.25\ in\cdot lbf)(1100\ RPM)}{63000}$$

$$\boxed{P = 0.02182\ hp}$$

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# CALCULATING / Design For Input Shaft

Knowns:

- $T_{in} = 0.46875$ in-lbf
- $P = 0.02182$ hp
- $N_{in} = 2933.33$ RPM
- GR = 24:9
- $N_{out} = 1100$ RPM
- $T_{out} = 1.25$ in-lbf
- $\varnothing_{input} = 0.125$ in
- $\varnothing_{output} = 0.25$ in

- From M.H.
  Max Shear Strength For power Transmitting Shaft (Input shaft) = 4000 psi
  ($\tau_{max\ input} = 4000$ psi)
  (Max Shear Stress For short shaft (output) = 8500 psi)
  ($\tau_{max\ output} = 8500$ psi)

Unknowns:

- $\tau_{input}$ (Torsional shear stress Input)
- $\tau_{output}$ (Torsional Shear stress output)

Torsional Shear Stress Input: (EQ2)

$$\tau_{input} = \frac{Tc}{J} = \frac{T(\frac{d}{2})}{(\frac{\pi}{32}d^4)} = \frac{16T}{\pi d^3} = \frac{16T_{in}}{\pi \varnothing_{input}^3}$$ (EQ1)

$$\tau_{input} = \frac{16(0.46875\ in\cdot lbf)}{\pi(0.125\ in)^3} = 1222.31\ psi$$

$\tau_{input} < \tau_{max\ input}$

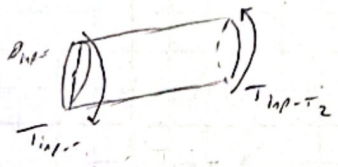
Torsional Shear stress output:

$$\tau_{output} = \frac{Tc}{J} = \frac{T(d/2)}{(\frac{\pi}{32}d^4)} = \frac{16T}{\pi d^3} = \frac{16T_{out}}{\pi \varnothing_{output}^3}$$

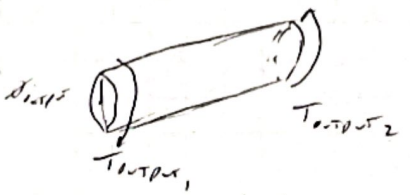
$$\tau_{output} = \frac{16(1.25\ in\cdot lbf)}{\pi(0.25\ in)^3} = 407.44\ psi$$

$\tau_{output} < \tau_{max\ output}$

FBD Input & output:




$\Sigma T_{input} = 0 = T_{input_1} - T_{input_2}$




$\Sigma T_{output} = 0 = T_{output_1} - T_{output_2}$

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# Gear Force Calculations:

Known: (From McMaster-Carr)

- $T_{OUT} = 1.25$ in·lbf
- $T_{in} = 0.46875$ in·lbf
- $D_{OUT} = 1.00$ in (pitch diameter) (Gear)
- $D_{in} = 0.375$ in (pitch diameter) (pinion)
- $\theta = 20°$

Unknowns:

- $F_T$ (Tangential Force)
- $F_r$ (radial Force)

Solving For $F_T$ (input : output):

$$F_T = F_{T_{in}} = F_{T_{OUT}} = \frac{2T_{OUT}}{D_{OUT}} = \frac{2T_{in}}{D_{in}} \quad \text{(eq 1)}$$

$$\frac{2(1.25\ \text{in·lbf})}{1.00\ \text{in}} = \frac{2(0.46875\ \text{in·lbf})}{0.375\ \text{in}}$$

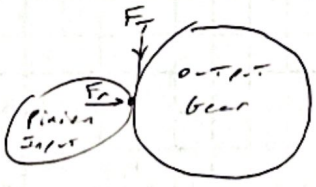
$$\boxed{2.5\ \text{lbf} = 2.5\ \text{lbf}}$$

Solving For $F_r$:

$$F_r = F_T(\tan(\theta)) \quad \text{(eq 1)}$$

$$F_r = 2.5\tan(20) = \boxed{0.90993\ \text{lbf}}$$

FBD:




Resultant Force ($F_R$):

$$F_R = \sqrt{(F_T)^2 + (F_r)^2}$$

$$F_R = \sqrt{(2.5)^2 + (0.90993)^2}$$

$$\boxed{F_R = 2.6604\ \text{lbf}}$$

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# Journal Bearings

## Shaft Surface Velocity:

Knowns:

- $\varnothing_{input} = 0.125$ in
- $\varnothing_{output} = 0.25$ in
- $N_{input} = 2933.3$ RPM
- $N_{output} = 1100$ RPM
- $T_{in} = 0.46875$ in-lbf
- $T_{out} = 1.25$ in-lbf

Unknowns:

- $V_{in}$ $\left(\frac{ft}{min}\right)$
- $V_{out}$ $\left(\frac{ft}{min}\right)$

General Equation: $V = \pi d \left(\frac{in}{rev}\right) \cdot N\left(\frac{rev}{min}\right) \cdot \left(\frac{1 ft}{12 in}\right)$

$$\boxed{V = \frac{\pi d (N)}{12}} \quad (eq\ 1)$$

Input Shaft:

$$V_{in} = \frac{\pi \varnothing_{in} (N_{in})}{12} = \frac{\pi (0.125 in)(2933.3\ RPM)}{12}$$

$$\boxed{V_{in} = 95.99\ ft/min}$$ Max Surface Velocity

Output Shaft:

$$V_{out} = \frac{\pi \varnothing_{out} (N_{out})}{12} = \frac{\pi (0.25 in)(1100 RPM)}{12}$$

$$\boxed{V_{out} = 71.99\ ft/min}$$

## Calculating Bearing Pressure:

Knowns:

- $\varnothing_{input} = 0.125$ in
- $\varnothing_{output} = 0.25$ in
- $F_R = 2.6604$ lbf
- (Assumption) $L_{input} = 1.25$ in
- (Assumption) $L_{output} = 1.25$ in
- $F_T = 2.5$ lbf
- $F_r = 0.90993$ lbf

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# Calculating Bearing Pressure (cont.)

Unknowns: • $P_{in}$ (pressure on Input)
• $P_{out}$ (pressure on output)

Calculating Input Bearing Pressure ($P_{in}$):

$$P_{in} = \frac{F_R}{\emptyset_{in} \cdot L_{in}} = \frac{2.6604\,lb_f}{(0.125\,in)(1.25\,in)}$$

(EQ 1)

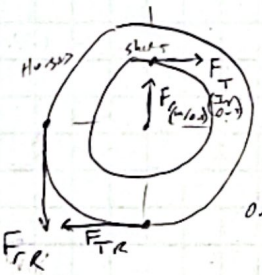
$$\boxed{P_{in} = 17.0266\,psi}$$ max pressure

Calculating Output bearing pressure ($P_{out}$):

$$P_{out} = \frac{F_R}{\emptyset_{out} \cdot L_{out}} = \frac{2.6604\,lb_f}{(0.25\,in)(1.25\,in)}$$

$$\boxed{P_{out} = 8.5133\,psi}$$

FBD:




(Same For Both Input & Output Force Locations)

(Reaction Forces)

(Resultant Force of Reaction Forces = $F_{RR}$)

Output/Input: $\Sigma F_x = 0 = F_T - F_{TR}$

$$\boxed{F_T = F_{TR} = 1.25\,lb_f}$$

$\Sigma F_y = 0 = F_r - F_{rR}$

$$\boxed{F_r = F_{rR} = 0.4551\,lb_f}$$

Solving For PV:

Knowns:
- $P_{in} = 17.0266\,psi$
- $V_{in} = 95.99\,ft/in$
- $P_{out} = 8.5133\,psi$
- $V_{out} = 71.99\,ft/min$

Unknowns:
- $PV_{in\,max}$
- $PV_{out\,max}$

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Solving For PV (cont.)

$PV_{INPUT} = P_{in}(V_{in})$ (cell 1)

$= (17.0266 psi)(95.99 FT/min)$

$PV_{INPUT} = 1634.3833 \; psi \cdot FT/min$

MAX PV

$PV_{OUTPUT} = P_{OUT}(V_{OUT})$ (cell 2)

$= (8.5133 psi)(71.99 FT/min)$

$PV_{OUTPUT} = 1612.8724 \; psi \cdot FT/min$

Material Selection:

Porous Bronze:

load capacity: 4500 psi > $P_{in}$ : > $P_{out}$

Speed limit: 1500 FT/min > $V_{in}$ : > $V_{out}$

PV: 50 000 psi · FT/min > $PV_{in}$ : > $PV_{out}$

Solving For Thickness of Journal Bearing:

Knowns:
- $L_{in}$ = 1.25 in (Assumption)
- $L_{out}$ = 1.25 in (Assumption)
- $F_R$ = 2.6604 lbf
- $F_b$ =
- $\sigma_{UTS}$ (Ultimate Tensile Strength) = 14000 psi (Design Hoop)
- FOS = 4 (Design Choice)

Unknowns:
- $Th_{min}$ (Thickness of Journal Bearing Min. Needed)

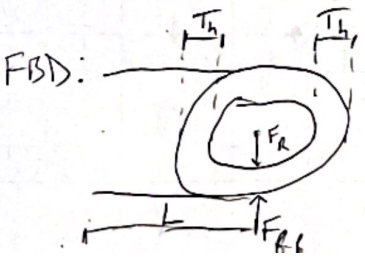
Solving For $Th_{min}$: $\sigma_{allow} = \frac{\sigma_{UTS}}{FOS} = \frac{14000 psi}{4} = 3500 psi$

$\sigma_{allow} = \frac{F}{A} = \frac{F_R}{A} = \frac{F_R}{2Th_{min}}$

$\sigma_{allow} = \frac{F_R}{2Th} = \frac{\sigma_{UTS}}{FOS} \rightarrow Th_{min} = \frac{F_R(FOS)}{2L(\sigma_{UTS})}$

$Th_{min} = \frac{2.6604 lbf (4)}{2(0.25 in)(14000 psi)} = 0.001520 in$

FBD:




Resistant Force ($F_R$) = 2.6604 lbf

Reaction Resistant Force ($F_{RR}$) = $F_R$

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Solving For $\emptyset$ of Journal Bearing:

Values:

- $Th_{min} = 0.00076$ in
- $\emptyset_{in} = 0.125$ in
- $\emptyset_{out} = 0.25$ in

Unknowns: • $OD_{out}$ (Journal diameter output)

• $OD_{in}$ (Journal diameter input)

Solving For $OD_{out}$:

$$OD_{out} = 2(Th_{min}) + \emptyset_{out}$$

$$= 2(0.00152\text{in}) + (0.25\text{in})$$

$$\boxed{OD_{out} = 0.2530\text{4 in}}$$

Solving For $OD_{input}$:

$$OD_{input} = 2(Th_{min}) + \emptyset_{in}$$

$$OD_{input} = 2(0.00152\text{in}) + 0.125\text{in}$$

$$\boxed{OD_{input} = 0.1280\text{4 in}}$$

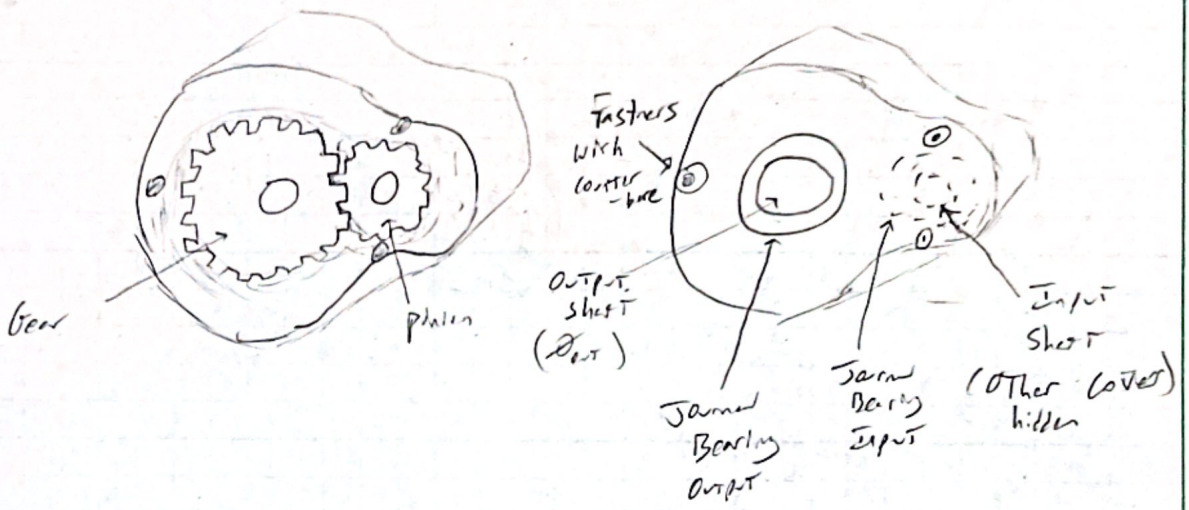
Journal Bearings on McMaster-Carr

output Journal Bearing: # 6391K131

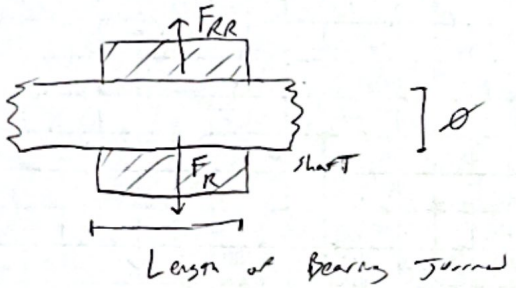
Input Journal Bearing: # 6391K111

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Sketch:




FBD: Pressure Bearing: Velocity:




Solving For CD:

$$C = \frac{\left[\left(\frac{N_{pinion}}{D_{pitch}}\right) + \left(\frac{N_{Gear}}{D_{pitch}}\right)\right]}{2}$$

$$C = \frac{\left[\frac{18\,in}{48\,in} + \frac{48\,in}{48\,in}\right]}{2} = \boxed{0.6875\,in}$$